

International University

Data Science and Data Visualization

Lab 02 — D3.js: From Data to Visualization

Work mode: Groups of 2

In-class duration: 2.5 hours (150 minutes)

Homework: 1 week, up to 3 hours per group

Points: 100

Main topic: D3 data, data binding, dynamic visual encoding, CSV loading, visualization verification

1. Learning Objectives

By the end of this lab, you should be able to:

LO1 — Explain Data Binding

Explain how D3 connects data to DOM elements using:

- `selectAll()`
- `data()`
- `enter()`
- `append()`
- datum `d`

LO2 — Map Data to Visuals

Implement a D3 visualization in which data values determine visual properties such as:

- position;
- height;
- color;
- text;
- size.

LO3 — Load and Verify Data

Load CSV data with D3, handle numeric values correctly, and verify that the visualization represents the data correctly.

LO4 — Debug and Adapt

Find and fix a visualization defect and modify an existing visualization when the dataset or requirement changes.

LO5 — Explain and Own the Solution

Explain the important parts of your pair's solution and make a small modification during the Group Check.

Main learning bottleneck: Given a dataset and D3 code, can you explain exactly how each datum becomes a visual element?

2. Lab Flow

THINK → BUILD → VERIFY → ADAPT → GROUP CHECK

AI OFF AI ON AI ON AI ON OWNERSHIP

Phase	Time	AI
THINK	25 min	OFF
BUILD	60 min	ON
VERIFY	25 min	ON
ADAPT	20 min	ON
GROUP CHECK	15 min	Instructor-defined
Buffer / transition	5 min	—
Total	150 min	

The objective is **not** to prevent AI use.

The objective is:

Think first. Ask AI second. Verify always. Own the result.

3. Work in Groups of 2

You will work with one partner.

Use the pair as a reasoning unit:

Discuss

↓

Decide

↓

Build

↓

Review

↓

Test

↓

Switch roles

Suggested roles

Driver

Writes code and runs the program.

Navigator / Reviewer

- discusses the problem;
- checks assumptions;
- asks “Why?”;
- reviews AI output;
- proposes tests.

Switch roles during the lab.

Do not work like this:

Student A → writes everything
Student B → watches

4. AI Policy

THINK — AI OFF

Do not use:

- ChatGPT;
- Claude;
- GitHub Copilot;
- AI coding assistants.

You may:

- discuss with your partner;
- ask the instructor to clarify requirements;
- use the lab handout;
- use your own notes.

BUILD / VERIFY / ADAPT — AI ON

AI is allowed.

You may use AI to:

- explain D3 syntax;
- suggest implementation approaches;
- debug;
- review code;
- identify edge cases;
- generate boilerplate.

However:

Do not submit code that neither member can explain.

5. Scenario

You are building a small web page for an instructor who wants to understand student performance.

The page will contain several D3 visualizations:

1. a dynamic bar chart;
2. a scatterplot of midterm and final exam scores;
3. a histogram of numerical values.

The instructor may later:

- replace the dataset;
- change the number of values;
- change a visual encoding;
- change a rule such as the passing threshold.

Therefore, the visualizations should not work only for the original example data.

6. Setup — 10 Minutes

Create a project such as:

```
lab02-d3/  
  index.html  
  script.js  
  style.css
```

Use **D3 only** for visualization.

Reference:

- D3: <https://d3js.org/>
- D3 tutorials: <https://github.com/d3/d3/wiki/Tutorials>
- JSON reference: https://developer.mozilla.org/en-US/docs/Web/JavaScript/Reference/Global_Objects/JSON

Read the assigned material from *Interactive Data Visualization for the Web*, especially the data and data-binding concepts.

Run your page through a local web server rather than relying on `file:///`.

7. THINK — AI OFF — 25 Minutes

7.1 Predict Before You Run

Consider:

```
var dataset = [10, 20, 30];
```

```
d3.select("svg")  
  .selectAll("rect")  
  .data(dataset)  
  .enter()  
  .append("rect");
```

Without running the code, answer:

1. How many `rect` elements will be created?
2. What value is bound to the first `rect`?
3. What value is bound to the third `rect`?
4. What does `d` represent in:

```
.attr("height", function(d) {  
  return d;  
})
```

5. What would happen if `dataset` contained 5 values?
-

7.2 Understand the D3 Pipeline

Complete this diagram:

```

dataset
  ↓
-----
  ↓
data()
  ↓
-----
  ↓
append("rect")
  ↓
visual elements

```

Then explain the role of:

- `selectAll()`
- `data()`
- `enter()`
- `append()`

Do not copy a definition. Explain it in your own words.

7.3 Trace d

For:

```
var dataset = [10, 60, 30, 80];
```

Consider:

```
.text(function(d) {
  return d;
})
```

Complete:

Iteration	Value of d
1	
2	
3	
4	

Then answer:

Why can D3 use the same function for all four elements?

7.4 Predict Visual Encoding

Consider:

```
.style("fill", function(d) {
  if (d > 50) {
    return "red";
  }
  return "black";
});
```

For:

[10, 60, 30, 80]

predict which elements will be red.

Then answer:

What is the mapping rule?

data value → -----

7.5 CSV Reasoning

Suppose a CSV contains:

Name,Midterm,Final

A,80,70

B,40,55

Before coding, answer:

1. What type might **Midterm** have after CSV loading?
2. What type might **Final** have?
3. Why could this matter when calculating:

$0.4 \times \text{Midterm} + 0.6 \times \text{Final}$

4. What should a row-conversion function accomplish?
-

7.6 THINK Checkpoint

Before using AI, your pair must have:

- [] Explained `selectAll/data/enter/append`
- [] Traced `d` through a dataset
- [] Predicted a color mapping
- [] Explained a CSV type issue
- [] Explained the `data → visual` pipeline

Show your answers to the instructor.

After the checkpoint, AI is allowed.

8. BUILD — AI ON — 60 Minutes

You will implement three visualizations.

The implementation details are intentionally left partly open.

Your pair must decide how data maps to visual properties.

9. Part A — Dynamic Bar Chart

The original task requires a bar chart that can draw an arbitrary dataset, with value-based coloring and value labels.

Requirements

Create a reusable visualization function conceptually similar to:

`drawBarChart(data)`

Your chart must:

- accept a dataset;
- create one visual bar for each data value;
- use the data value to determine bar height;
- use the data value to determine color;
- display the value as a label;
- work when the number of values changes.

Create an initial dataset containing 20 random values.

You decide:

- SVG dimensions;
- bar width;
- spacing;
- color rule;
- height scaling.

Do not hard-code 20 bars individually.

Required reasoning

Your pair must be able to explain:

```
data[i]
  ↓
bar i
  x position
  height
  color
  label
```

Quick test

After your first version works, replace the dataset with:

`[7, 21, 3, 45, 12, 90, 4]`

Then replace it with a larger dataset.

Does the chart still work?

Record what you had to change.

10. Part B — Scatterplot

Use the provided CSV source:

<https://tungh.github.io/data/vis-lab2-data.csv>

Read the CSV with D3.

Your scatterplot must show:

- midterm score on the x-axis;
- final exam score on the y-axis;
- one circle per student;
- radius = 5 pixels.

Calculate each student's course score as:

```
course score =  
  0.4 × midterm  
+ 0.6 × final
```

Highlight students whose course score is below 50.

Important

Do not assume that values read from CSV are already numeric.

Verify the data before using it in calculations.

Your pair must explain

```
CSV row  
↓  
student object  
↓  
midterm / final  
↓  
x / y position  
↓  
course score  
↓  
pass/fail visual encoding
```

11. Part C — Histogram

Use the same numerical data from the bar-chart part and create a histogram with **10 bins**.

Your pair must be able to explain:

1. What does one bin represent?
2. Why is a histogram different from the original bar chart?
3. What happens when the data range changes?
4. Why might the number of observations in each bin change?

Do not manually create ten unrelated bars.

The histogram must be derived from the data.

12. AI Collaboration Record

Record at least **3 meaningful AI interactions per pair**.

#	What we asked	What AI suggested	What we accepted/changed	How we verified
1				
2				
3				

At least one interaction must be a **review or verification** request.

Good:

Review our bar-chart data binding. Identify possible errors or assumptions. Do not rewrite the code.

Weak:

Write the whole lab for us.

AI use is allowed. Blind acceptance is not.

13. VERIFY — 25 Minutes

A visualization that appears on screen is not necessarily correct.

Your pair must perform the following verification tasks.

13.1 Data Count Test

For:

[10, 20, 30, 40, 50]

verify:

How many visual elements should exist?

Check the DOM.

13.2 Data Mapping Test

Verify that:

first data value → first visual element

second data value → second visual element

...

Use the browser developer tools if necessary.

13.3 Edge Dataset Tests

Test your bar chart with:

Test A`[10]`**Test B**`[10, 20, 30]`**Test C**`[0, 0, 0]`**Test D**`[5, 100, 20, 80]`**Test E**`[]`

Record:

Dataset	Expected	Actual	PASS/FAIL
[10]			
[10,20,30]			
[0,0,0]			
[5,100,20,80]			
[]			

14. VERIFY — Debugging Challenge

Your instructor may give your pair a visualization containing a small defect.

Possible defects include:

Bug A — Missing datum

The dataset contains 5 values but only 4 visual elements appear.

Bug B — Wrong type

A CSV numeric field is still a string.

Bug C — Wrong pass rule

A student with course score exactly 50 is incorrectly highlighted as failing.

Bug D — Wrong mapping

The chart uses the index instead of the data value for a visual property.

Bug E — Wrong assumption

The visualization works for 20 values but breaks when the dataset contains 7 values.

Your pair must follow:

```
Observe
↓
Identify symptom
↓
Explain likely cause
↓
Fix
↓
Retest
```

Do not immediately ask AI for the final fix.

First write:

Our hypothesis: _____

Then you may use AI to review your reasoning.

15. VERIFY — AI Review

Ask AI to review one important part of your visualization.

Suggested prompt:

Review this D3 code. Identify possible data-binding errors, incorrect assumptions, or edge cases.
Do not rewrite the code.

Then answer:

1. What did AI identify correctly?
2. What did AI miss?
3. Did AI suggest anything irrelevant?
4. Which suggestion did you verify?
5. What did you change?

16. ADAPT — 20 Minutes

The instructor now changes the requirement.

Change 1 — Bar Chart

The client says:

Highlight every value above the dataset average.

Your chart should now distinguish:

```
value <= average
value > average
```

Do not hard-code the threshold.

The threshold must be calculated from the current dataset.

Test at least:

[10, 20, 30, 40]

and another dataset of your choice.

Change 2 — Scatterplot

The instructor changes the passing threshold:

A student is considered failing if the course score is **below 60** instead of below 50.

Modify the visualization.

Do not rewrite the entire chart.

Your task is to identify which part of the data-to-visual mapping must change.

Adapt Reflection

Answer:

1. Which change was easier?
 2. Why?
 3. Which part of your original design made the change easy or difficult?
 4. What would you refactor if more visual rules were added later?
-

17. GROUP CHECK — 15 Minutes

The instructor will select one or two short challenges for your pair.

Possible challenges:

C01 — Explain d

What does `d` represent in this line?

```
.attr("height", function(d) {  
    return d;  
})
```

C02 — Explain `enter()`

Why is `enter()` needed in this code?

C03 — Predict

Given a dataset, predict the number of SVG elements.

C04 — Debug

Find a small data-binding or CSV conversion bug.

C05 — Modify

Change the bar-chart rule from:

above average

to:

above median

C06 — Explain CSV

Why should numeric CSV fields be converted before numerical calculations?

C07 — Scatterplot

Explain how one student's midterm/final values become a point.

C08 — Histogram

Explain what one bin represents.

Both members should participate.

The purpose is:

Demonstrate ownership of your solution.

18. Submission

Submit one shared submission per pair.

```
lab02-d3/
  index.html
  script.js
  style.css
  other required files
```

Also submit:

1. THINK answers;
 2. AI Collaboration Record;
 3. VERIFY test record;
 4. debugging result;
 5. ADAPT reflection.
-

19. Assessment — 100 Points

Component	Points
THINK — data-binding reasoning	20
BUILD — D3 implementation	20
VERIFY — testing/debugging	20
ADAPT — requirement changes	15
GROUP CHECK — ownership	25
Total	100

Component	Points
-----------	--------

A working visualization alone is not sufficient for a high score.

20. Homework — One Week — Group of 2 — Maximum 3 Hours

The homework extends the same D3 visualizations.

H1 — Interactive Dataset Change

Add a simple control that allows the user to regenerate or replace the bar-chart dataset.

The visualization must update from the new data.

You must explain:

What happens when the data changes?

H2 — Second Visual Encoding

Modify the bar chart so that one additional visual property depends on the data.

Examples:

- opacity;
- label size;
- border;
- color intensity.

Explain the mapping rule.

H3 — Scatterplot Requirement Change

Add one additional visual encoding to the scatterplot.

For example:

Circle size represents the course score.

You must explain why the chosen encoding makes sense.

H4 — Verification

Create at least 5 meaningful tests.

Include at least:

- one normal case;
- one boundary case;
- one unusual dataset;
- one data-loading or data-type case;
- one requirement-change case.

H5 — AI Review

Ask AI to review one important visualization function.

Record:

- what AI found;
 - what AI missed;
 - what you verified;
 - what you changed.
-

21. Homework Deliverables

Submit:

1. Updated source code

The working D3 visualization.

2. Short Design Note

Maximum 1 page:

1. What did we change?
2. What data-to-visual mapping did we add?
3. Why did we choose it?
4. What was difficult?
5. What did AI help with?
6. What did we verify ourselves?
7. What would we improve?

3. Test Record

At least 5 meaningful tests.

4. AI Review Record

At least 2 meaningful AI interactions.

22. Final Reflection

Discuss with your partner:

What did AI help us do faster?

What did AI not decide for us?

Which D3 concept did we understand better after this lab?

Which bug or edge case did we discover?

Could we explain how one datum becomes one visual element?

Could we modify our visualization without rebuilding everything?

Core Principle

Think first. Ask AI second. Verify always.

AI can generate D3 code. You must understand the data, the mapping, the verification, and the change.